TeachDB: A Conceptual Architecture for Growing AI Through Accumulated Human Reasoning

Abstract: *Current AI development focuses largely on the accumulation of information. This proposal introduces TeachDB, a conceptual architecture designed to grow AI systems by accumulating human reasoning through continuous human–AI dialogue. By transitioning from static data storage to the active synthesis of reasoning patterns, TeachDB aims to provide a methodology for AI to evolve alongside a human partner, internalizing their unique cognitive frameworks over time.*

1. Introduction

Modern AI architectures typically optimize for the retrieval and processing of static information. While effective for short-term tasks, these systems lack the capacity to grow in cognitive sophistication through persistent engagement. This research proposes TeachDB as an architectural meta-layer that prioritizes the accumulation of reasoning, enabling AI systems to mature into personalized cognitive engines.

2. The Methodology of Growing AI

TeachDB is not a replacement for existing technologies such as RAG, Knowledge Graphs, or Agentic Memory. Instead, it proposes a higher-level orchestration layer that utilizes these components to synthesize reasoning. The core hypothesis is that by treating reasoning—the logical path taken to reach a conclusion—as the fundamental unit of growth, AI can develop an internal model that reflects the intellectual style and heuristics of its human collaborator.

3. System Architecture

TeachDB functions as a persistent feedback loop designed to iteratively expand the AI's reasoning model:

Human → Continuous Dialogue → Reasoning Extraction Engine → Reasoning Graph Update → TeachDB (Orchestration Layer) → Knowledge Evolution Engine → LLM Synthesis → Future Dialogue

The *Reasoning Extraction Engine* maps dialogue into formal logical structures, while the *Knowledge Evolution Engine* ensures these structures integrate into the system's growing cognitive model, allowing for long-term refinement rather than mere storage.

4. Case Study: Cognitive Internalization

Consider an AI partner working with a novelist. Rather than training on text samples, TeachDB maps the underlying logical connections and thematic axioms of the author’s work. Over time, the AI develops a reasoning model that can simulate the author's deductive approach to novel intellectual queries (e.g., AGI), generating insights that are structurally congruent with the author’s unique style of thinking, even in domains they never explicitly discussed.

5. Comparison and Integration

Technology	Role in TeachDB
RAG/GraphRAG	Provides the source evidence for reasoning.
Vector DB	Maintains the latent memory pool.
TeachDB	**Synthesizes and grows the reasoning model.**

6. Research Questions

- To what extent can formal reasoning graphs capture individual cognitive heuristics?
- How does continuous dialogue facilitate the growth of an AI's internal reasoning model?
- What are the metrics for evaluating the "growth" of an AI’s thinking capability?

7. Limitations and Ethics

This architecture faces significant challenges regarding cognitive provenance, structural scalability, and the ethics of mirroring individual thought patterns. Future research must address how to supervise the "growth" of such systems to ensure alignment and intellectual integrity.